

MERCHANT TAXONOMY AUDIT + MASTER FRAUD INTELLIGENCE DOCUMENT

PART 3 – FINAL MERCHANT TAXONOMY

FAMILY MCH-001: AI-GENERATED SYNTHETIC MERCHANT ONBOARDING

A. IDENTITY

Field	Value
Attack ID	MCH-001
Attack Family	AI-Generated Synthetic Merchant Onboarding
Attack Subtypes / Variants	Synthetic business fraud; Fraudulent merchant onboarding; Merchant impersonation; Merchant misrepresentation / MCC fraud; Synthetic UBO/director fraud; Hybrid synthetic (real + fabricated data)
Primary Lifecycle Stage	Merchant (Stage 6 – Onboarding)
Secondary / Cross-stage Stages	Identity/KYC (Stage 1 – synthetic identity creation); Account Creation (Stage 2); Acquirer (Stage 7)
Target / Actor	Merchant onboarding workflows; KYB verification; Underwriting; Acquirer risk assessment
Attack Objective	Obtain fully verified merchant accounts under false pretenses to process fraudulent transactions, launder money, access credit facilities, or resell accounts on fraud marketplaces
Merchant Type / Context	All merchant types; particularly e-commerce, digital goods, and high-risk MCCs

B. FRAUD UNDERSTANDING

What is the attack?

Attackers use generative AI to create entirely fictitious business entities – complete with fabricated company structures, synthetic directors/UBOs, and internally consistent document sets – that pass KYB verification and obtain legitimate merchant accounts. The fraud industry has moved from fake IDs to fake business files because a fabricated business yields trade credit, a merchant account, loan facilities, and a node in a laundering network. Criminal networks now use synthetic businesses, AI-generated identities, and agent-driven automation to exploit weak links in onboarding.

Traditional mechanism:

Before GenAI, fraudulent merchant onboarding relied on stolen or forged physical documents (Photoshop, physical counterfeiting), stolen business identities from data breaches, manual fabrication of business details, and template-based document forgery with detectable artifacts. Each application required significant manual effort; scalability was limited.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Document quality	Detectable artifacts, inconsistent details	Complete, internally consistent business files
Scalability	Manual, dozens per campaign	Industrial-scale, thousands simultaneously
Business entities	Stolen or paper companies	Synthetic companies with no real existence
Detection evasion	Detectable by document forensics	Evades visual and template validation
Speed	Weeks per application	Minutes per application

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- Generate complete, internally consistent business document sets
- Create synthetic companies with no real-world existence
- Scale from dozens to thousands of fraudulent applications

- Produce documents that pass visual and template validation
- Construct fictitious company structures with supporting registries

HARD GenAI Removal Test result: PASS — GenAI creates a qualitatively new capability. The transformation from manual, detectable document forgery to industrial-scale synthetic business creation with perfect internal consistency is a qualitative leap. KYB's instinctive cross-document consistency check is a test that generated files pass by construction. The forged file is often MORE internally consistent than a real one.

Prerequisites:

- Access to generative AI tools for document generation (LLMs, image generation)
- Understanding of KYB verification logic and document requirements
- (Optional) Breached business data for hybrid synthetic identities
- Ability to submit applications through merchant onboarding flows

Attack Flow:

1. Attacker identifies target KYB requirements (document types, verification workflow)
2. Attacker selects generative AI tools for document production
3. Attacker creates synthetic business entity with fabricated company structure, UBOs, directors
4. AI generates complete document set: incorporation certificates, bank statements, financial statements, invoices
5. Application submitted through merchant onboarding flow
6. KYB verification processes the synthetic entity and documents
7. Account provisioned with merchant ID (MID)
8. Account used for fraudulent transactions, laundering, or resale

Merchant Lifecycle Point: FRAUD ENTERS AT MERCHANT ONBOARDING — the fraudulent merchant identity and documents are submitted during onboarding. The synthetic entity is created specifically to obtain a merchant account.

Downstream Payment / Financial Consequence:

- Fraudulent transactions processed through synthetic merchant accounts
- Chargebacks impact payment network
- Funds laundered through mule accounts
- Credit facilities accessed and defaulted (bust-out)
- Merchant accounts resold on fraud marketplaces
- Regulatory penalties for acquirer
- 15% of company registration certificates submitted for corporate account opening are fake
- One in three documents analyzed in 2025 showed structural tampering

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Merchant Intelligence Required?
Merchant identity	Business name, registration number, UBO details	Synthetic entities have fabricated or inconsistent details	Entity resolution; Registry lookups	Yes
Merchant profile	MCC, business type, industry	May be inconsistent with actual activity	Rules + ML; Anomaly detection	Yes
Document	Document artifacts, metadata	AI-generated document patterns	Document forensics (CNN/Transformer)	Yes
Transaction	Volume, amounts, velocity, authorization rates	Anomalies indicate synthetic/fraudulent activity	ML anomaly detection	Yes
Customer relationship	Customer geography, repeat patterns	Inconsistent with declared business	Rules + ML	No
Device	Device fingerprint during onboarding	Automation/bot patterns	Device fingerprinting	Yes
Session	Application completion speed	Superhuman speed indicates automation	Temporal anomaly detection	No
Behavioral	Form-filling patterns	Unnatural patterns indicate automation	Behavioral analytics	No
Network/graph	Shared attributes across merchants	Synthetic identity clusters	Graph Neural Network	Yes — essential
Temporal	Application timing, velocity	Off-hours, rapid succession	Velocity rules + ML	Yes

Evasion Considerations (Defensive Threat Modeling):

An adaptive attacker might attempt to:

- Use unique document patterns and generators per application to avoid fingerprinting
- Vary application timing and patterns (not always off-hours)
- Create minimal but convincing digital footprints (websites, social presence)
- Use genuine business data fragments combined with fabricated elements
- Submit applications during legitimate business hours
- Rotate IP addresses, devices, and other infrastructure
- Adapt document quality based on detection feedback

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
New legitimate business	No transaction history; limited data	Blocking legitimate startups
Thin-file business	Limited digital footprint	Blocking legitimate new businesses
Rapid legitimate growth	Volume spikes; new customers	Blocking successful merchants
Cross-border business	Unusual geography	Blocking legitimate international sales
Home-based business	PO box; limited physical presence	Blocking legitimate small businesses
Professional design	"Too perfect" documents/images	False positives on professional materials

D. SIMULATION INTELLIGENCE

Field	Value
Simulation Type	Hybrid (Algorithmic + Agentic)
Generation Parameters	Jurisdiction; business type; document quality; cultivation duration; document artifact patterns
Synthetic Merchant	Fictitious business entity with fabricated company structure
Business profile	Declared revenue, employee count, years in operation, product/service description
Merchant category	Synthesized MCC (may be low-risk)
Merchant attributes	Legal name, registration number, jurisdiction, business type, industry, location
Synthetic customers	Synthetic customer profiles (may be used for transaction laundering)
Merchant devices/sessions	Device fingerprint during onboarding; IP geolocation; session timing
Synthetic transactions	Cultivation-phase transactions (low volume); fraud-phase transactions (high volume, laundering)
Refund/chargeback events	May be elevated in fraud phase
Merchant behavioral events	Onboarding flow completion (form filling, document upload, verification interaction)
Merchant relationships	Connections to other synthetic merchants (shared infrastructure, attributes)
Network relationships	Fraud ring clustering; mule network connections
Temporal Behavior	Onboarding timing; cultivation duration; fraud phase timing
GenAI-generated content	Business documents (incorporation certificates, bank statements, financial statements, invoices)
Attack Labels	Synthetic merchant; fraudulent onboarding; synthetic business fraud
Variant / Mutation Parameters	Jurisdiction; business type; document quality; cultivation duration; document artifact patterns; infrastructure rotation
Legitimate Counterpart	New legitimate businesses; thin-file businesses; startups
Realistic Edge Cases	Hybrid synthetic (real + fabricated data); partially verified accounts; accounts resold on marketplaces

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
AI-generated business documents are a major KYB fraud vector	Signzy	Signzy	2026	Industry report	VERIFIED
One in three documents analyzed in 2025 showed structural tampering	Resistant AI	Signzy	2026	Industry data	VERIFIED

Claim	Source	Publisher	Date	Evidence Type	Confidence
15% of company registration certificates submitted for corporate account opening are fake	Resistant AI	Signzy	2026	Industry data	VERIFIED
Generative tools create hyper-realistic fake documents, synthetic identities, deepfake photos	LegitScript	LegitScript	2025	Webinar analysis	VERIFIED
AI agents attempt thousands of onboarding flows simultaneously	Duna	Fintech.global	2026	Industry analysis	VERIFIED
Fake merchant onboarding is no longer manual; AI-generated identities pass static KYC checks	Stripe	LinkedIn/Fintech Wrap Up	2025	Industry report	VERIFIED
AI-generated IDs pass visual and template validation	Stripe	LinkedIn/Fintech Wrap Up	2025	Industry report	VERIFIED
Synthetic entity fraud shifted from niche to mainstream risk	DataBreachToday	DataBreachToday	2025	Industry analysis	VERIFIED
Synthetic businesses created for <\$150 with >\$100,000 payouts	DataBreachToday	DataBreachToday	2025	Industry analysis	VERIFIED
GenAI increasing speed, scale, effectiveness of synthetic-identity fraud	Boston Fed	Black Hat	2025	Government analysis	VERIFIED
Synthetic identity fraud increased eightfold in 2025, 11% of all fraud	LexisNexis	ValidSoft	2026	Industry report	VERIFIED
Mastercard fraud ring exploited onboarding flow; cut 20% of marketplace revenue	Trulioo/Mastercard	Trulioo	2025	Webinar case study	VERIFIED

F. PRODUCT MAPPING

Component	Implementation
Generate	Algorithmic generation of synthetic business entities with consistent attributes; AI-generated document sets with varying artifact patterns; Agentic generation of onboarding flow interactions
Defend	Document forensics (CNN/Transformer); Entity resolution graph; Registry verification; UBO identity verification; Behavioral biometrics for onboarding
Detection Approach	Supervised ML for document authenticity; Graph Neural Network for entity resolution; Anomaly detection for application patterns; Ensemble risk scoring
Mitigation	Allow (verified legitimate); Enhanced review (additional verification); Hold (delay onboarding pending manual review); Block (reject application; flag in consortium intelligence)
Evaluation	Precision; Recall; F1; AUC; False Positive Rate; Time-to-Detection

FAMILY MCH-002: AI-GENERATED SHELL / FACADE MERCHANT FRAUD

A. IDENTITY

Field	Value
Attack ID	MCH-002
Attack Family	AI-Generated Shell / Facade Merchant Fraud
Attack Subtypes / Variants	Front merchant fraud; Transaction laundering fronts; Shell company networks; Merchant collusion (as a technique); Commercial laundering (laundering as a service)
Primary Lifecycle Stage	Merchant (Stage 6 – Ongoing Payment Acceptance)
Secondary / Cross-stage Stages	Merchant onboarding (Stage 6); Payment Initiation (Stage 5); AML/Compliance (Stage 17); Acquirer (Stage 7)
Target / Actor	Merchant onboarding, acquirer monitoring, payment processing; The front merchant serves as a laundering vehicle
Attack Objective	Process illicit transactions (payments for prohibited goods/services, money laundering) through a legitimate-appearing merchant account that appears normal to the acquirer
Merchant Type / Context	E-commerce merchants; Digital goods; High-volume merchants; Cross-border merchants

B. FRAUD UNDERSTANDING

What is the attack?

Attackers use AI-generated front websites and merchant profiles to create convincing front businesses that process payments for illicit activity – laundering money through the payment system. Transaction laundering has evolved, with fraudsters able to deploy AI-built "front" websites so convincing that they evade standard onboarding detection. AI-powered design platforms now create sophisticated sites that appear legitimate, enabling transaction launderers to establish convincing front businesses at unprecedented speed. This is considered the "single highest-consequence form of merchant fraud" for acquirers.

Traditional mechanism:

Before GenAI, transaction launderers created basic "front websites" with telltale signs of being fraudulent (blurry images, clumsy spelling, broken layouts). Each front required manual effort to create and maintain. Detection was possible through obvious red flags.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Front quality	Basic, detectable	Sophisticated sites that appear legitimate
Speed	Months to establish	Days to build and scale
Scalability	Limited	Industrial-scale front creation
Detection evasion	Obvious red flags	Evades standard onboarding detection
Operation	Manual	Automated payment flows

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- Generate convincing front websites without design skills
- Create professional-looking sites that evade detection
- Scale front creation beyond manual capabilities
- Rapidly recreate fronts when detected

HARD GenAI Removal Test result: PASS – GenAI creates a qualitatively new capability. The transformation from detectable, basic front websites to indistinguishable, professional-grade sites that evade standard onboarding detection is a qualitative leap. Scam operations previously requiring months to establish can now be built and scaled in days using AI tools.

Prerequisites:

- Access to generative AI tools (website generation, image generation, copy generation)
- Ability to register a business entity (synthetic or real)
- Ability to obtain a merchant account
- Understanding of payment processing and laundering techniques
- Illicit business needing payment processing

Attack Flow:

1. Attacker identifies target industry for front business
2. Selects generative AI tools for website and content creation
3. Registers a business entity (may be synthetic)
4. AI generates professional-looking front website (product galleries, team pages, testimonials)
5. Attacker submits front merchant for onboarding
6. Merchant account provisioned based on convincing facade
7. Front website goes live; payment acceptance infrastructure enabled
8. Illicit transactions processed through the front merchant
9. Transactions appear legitimate (volume, velocity, amounts within expected ranges)
10. Funds laundered through shell merchant; shell may be abandoned and recreated

Merchant Lifecycle Point: FRAUD ENTERS AT MERCHANT ONBOARDING (the front is created for onboarding) AND CONTINUES THROUGH PAYMENT ACCEPTANCE (laundering transactions processed through the front merchant).

Downstream Payment / Financial Consequence:

- Funds laundered through shell merchant
- Chargebacks from undelivered goods/services (if consumer fraud involved)
- Regulatory penalties for acquirer
- Scheme violations (BRAM/VIRP)
- "Single highest-consequence form of merchant fraud" for acquirers

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Merchant Intelligence Required?
Merchant identity	Business details, registration	May be synthetic or inconsistent	Entity resolution	Yes
Merchant profile	Website quality, content	Template-heavy or AI-generated patterns	NLP; Image analysis	No
Transaction	Volume, velocity, patterns	Laundering indicators	ML anomaly detection	Yes
Customer relationship	Customer geography, patterns	Inconsistent with declared business	Rules + ML	No
Device	Device fingerprint	Infrastructure patterns	Device fingerprinting	Yes
Session	Session patterns	Automation indicators	Anomaly detection	No
Behavioral	Administrative behavior	Unnatural patterns	Behavioral analytics	No
Network/graph	Merchant connections	Shell networks / laundering networks	Graph Neural Network	Yes – essential
Temporal	Timing patterns	Off-hours, rapid changes	Velocity rules + ML	Yes
Website	AI-generated content	Image artifacts, text patterns	CNN/Transformer; NLP	No
Transaction laundering	Inconsistent descriptors	MCC vs. actual business	Rules + ML	Yes

Evasion Considerations:

An adaptive attacker might attempt to:

- Use varied AI tools and patterns across shells to avoid fingerprinting
- Rotate domains, IP addresses, and hosting providers rapidly
- Include some genuine transactions to build credibility
- Cloak content – showing different pages to crawlers vs. consumers
- Vary the sophistication of shells
- Use legitimate e-commerce platforms (Shopify, etc.) to host shells
- Abandon shells quickly before detection and recreate

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
New e-commerce business	New website, limited history	Blocking legitimate startups
Professional website design	"Too perfect" images	False positives on professional design
Small/home-based business	Limited physical presence	Blocking legitimate small businesses
Seasonal business	Volume spikes	Blocking legitimate seasonal patterns
New product line	Changes in business profile	Blocking legitimate business expansion

D. SIMULATION INTELLIGENCE

Field	Value
Simulation Type	Hybrid (Agentic + Algorithmic)
Generation Parameters	Industry; website quality; laundering method; volume; infrastructure patterns
Synthetic Merchant	Shell/facade business with AI-generated front presence
Business profile	Declared product/service, website URL, business description
Merchant category	Low-risk MCC (misrepresented)
Merchant attributes	Business name, registration, location, industry (may be synthetic)
Synthetic customers	Synthetic customers (may include genuine-looking transactions)

Field	Value
Merchant devices/sessions	Device fingerprint; IP geolocation; session patterns
Synthetic transactions	Mix of legitimate-looking and laundering transactions
Refund/chargeback events	May be elevated (undelivered goods)
Merchant behavioral events	Shell creation → onboarding → operation → laundering → abandonment
Merchant relationships	Connections to other shells (shared infrastructure)
Network relationships	Fraud ring clustering; laundering networks
Temporal Behavior	Shell creation timing; operation duration; abandonment patterns
GenAI-generated content	Front website (product galleries, team pages, testimonials, product descriptions)
Attack Labels	Shell merchant; facade merchant; transaction laundering front
Variant / Mutation Parameters	Industry; website quality; laundering method; domain rotation; content variation; infrastructure changes
Legitimate Counterpart	New e-commerce businesses; legitimate small businesses
Realistic Edge Cases	Partially legitimate shells (some genuine transactions); abandoned shells

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
AI-built front websites evade standard onboarding detection	The Financial Revolutionist	The FR	2025	Industry analysis	VERIFIED
Innocuous-looking storefronts generated by AI to mask illegal payment processing	The Financial Revolutionist	The FR	2026	Industry analysis	VERIFIED
Scam operations previously requiring months can now be built and scaled in days using AI	Ballerine	The Paypers	2026	Industry report	VERIFIED
Entire e-commerce websites generated in seconds for transaction laundering	LegitScript	LegitScript	2025	Webinar analysis	VERIFIED
Synthetic merchant sites with professional compliance docs, AI-generated every element	Visa	Benzinga	2025	Payment network report	VERIFIED
Transaction laundering is the "single highest-consequence form of merchant fraud"	Onlayer	Onlayer	2026	Industry guide	VERIFIED
AI-powered design platforms create sophisticated sites that appear legitimate	Forbes	Forbes	2025	News report	VERIFIED

F. PRODUCT MAPPING

Component	Implementation
Generate	Agentic generation of front websites with AI-generated content; Algorithmic generation of laundering transaction sequences; Network relationship generation
Defend	Website content authenticity scoring; Image forensics (CNN/Transformer); Graph Neural Network for laundering network analysis; Transaction anomaly detection; MCC consistency scoring
Detection Approach	CNN/Transformer for image authenticity; Graph Neural Network for laundering network analysis; Anomaly detection for transaction patterns; ML classification for MCC consistency
Mitigation	Allow (verified legitimate); Enhanced monitoring (watch for laundering patterns); Hold (delay onboarding pending review); Block (reject; flag in consortium intelligence)
Evaluation	Precision; Recall; AUC; False Positive Rate; Time-to-Detection

FAMILY MCH-003: AI-GENERATED FAKE MERCHANT STOREFRONTS

A. IDENTITY

Field	Value
Attack ID	MCH-003
Attack Family	AI-Generated Fake Merchant Storefronts

Field	Value
Attack Subtypes / Variants	Scam merchant storefronts; Brand impersonation stores; Fake e-commerce stores; Triangulation fraud; Credential/API abuse (as enabler); Fake product stores
Primary Lifecycle Stage	Merchant (Stage 6 – Onboarding / Payment Acceptance)
Secondary / Cross-stage Stages	Payment Initiation (Stage 5 – customer pays); Dispute/Chargeback (Stage 15 – non-delivery)
Target / Actor	Consumers, legitimate brands, payment networks
Attack Objective	Collect payments for goods never delivered (or counterfeit), harvest payment credentials, profit from consumer deception
Merchant Type / Context	E-commerce merchants; Consumer goods; High-demand products; Brand lookalikes

B. FRAUD UNDERSTANDING

What is the attack?

Attackers use generative AI to create professional-looking fake online storefronts that appear legitimate to consumers and payment systems. These storefronts have product galleries, "team" pages, five-star testimonials, and complete branding. A fraudster can now assemble a convincing brand lookalike in hours. Consumers worldwide lost an estimated \$442 billion to online scams in 2025. Fake shops rose 790% in Q1 2025.

Traditional mechanism:

Before GenAI, fake storefronts were template-based with obvious red flags: blurry photos, clumsy spelling, broken layouts. Each store required manual creation; scalability was limited; detection was possible through obvious quality issues.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Store quality	Blurry, broken layouts	Professional, studio-grade
Creation time	Days/weeks	Minutes/hours
Content	Generic, errors	Authentic-sounding reviews, perfect grammar
Scalability	Limited	Industrial-scale
Detection	Obvious red flags	Indistinguishable from legitimate

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to:

- Generate convincing storefronts without design skills
- Create professional-looking websites in minutes
- Scale fake storefront creation
- Produce studio-grade product images
- Generate synthetic content and fake reviews at scale

HARD GenAI Removal Test result: PASS – GenAI creates a qualitatively new capability. The transformation from detectable template-based fake stores to indistinguishable professional-grade storefronts is a qualitative leap. The tell-tale signs of a scam used to be obvious; today's generative AI removes those cues. Scammers no longer look like scammers.

Prerequisites:

- Access to generative AI tools (image generation, copy generation, website generation)
- Ability to create a merchant account or use payment acceptance
- Ability to attract customers (ads, social media, search)

Attack Flow:

1. Attacker identifies product category and target market
2. Uses GenAI to generate storefront, product images, descriptions, and reviews
3. Storefront appears professional with product galleries, team pages, testimonials
4. Attacker submits fake storefront for merchant onboarding
5. Account provisioned (may pass checks if storefront appears legitimate)
6. Fake storefront goes live
7. Customers make purchases believing storefront is legitimate
8. Payments processed through merchant account
9. Goods never delivered (or counterfeit)

10. Storefront abandoned; recreated under new domain

Merchant Lifecycle Point: FRAUD ENTERS AT MERCHANT ONBOARDING (fake storefront submitted) AND CONTINUES THROUGH PAYMENT ACCEPTANCE (customer purchases processed).

Downstream Payment / Financial Consequence:

- Consumers lose money for undelivered goods
- Chargebacks impact payment network
- Legitimate brands suffer reputational damage
- \$442 billion lost to online scams globally in 2025
- 20,000+ fake shop domains mapped

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Merchant Intelligence Required?
Merchant identity	Business details, registration	May be inconsistent	Entity resolution	Yes
Merchant profile	Website content, product images	AI-generated artifacts	CNN/Transformer; NLP	No
Transaction	Volume, velocity, patterns	Fraud indicators	ML anomaly detection	Yes
Customer relationship	Customer geography, patterns	May be inconsistent	Rules + ML	No
Device	Device fingerprint	Infrastructure patterns	Device fingerprinting	Yes
Session	Session patterns	Automation indicators	Anomaly detection	No
Behavioral	Administrative behavior	Unnatural patterns	Behavioral analytics	No
Network/graph	Storefront connections	Fake store networks	Graph Neural Network	Yes – essential
Temporal	Timing patterns	Promotional timing; rapid creation	Velocity rules + ML	Yes
Website	AI-generated content	Image artifacts, text patterns	CNN/Transformer; NLP	No
Refund/chargeback	High refund/chargeback rate	Non-delivery indicator	Rules + ML	Yes

Evasion Considerations:

An adaptive attacker might attempt to:

- Use varied AI tools and patterns across storefronts to avoid fingerprinting
- Rotate domains, IP addresses, and hosting providers rapidly
- Include some genuine transactions to build credibility
- Use cloaking to show different content to crawlers
- Host on legitimate e-commerce platforms
- Use social media ads to drive traffic before detection
- Abandon storefronts quickly and recreate

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
New legitimate e-commerce store	New website, limited history	Blocking legitimate startups
Professional product photography	"Too perfect" images	False positives on professional design
Small business with limited presence	May look "too new"	Blocking legitimate small businesses
Legitimate promotional campaign	Volume spike	Blocking legitimate promotions
Legitimate brand with strong design	Professional appearance	False positives on legitimate brands

D. SIMULATION INTELLIGENCE

Field	Value
Simulation Type	Hybrid (Agentic + Algorithmic)
Generation Parameters	Product category; storefront quality; impersonation type (brand clone vs. new brand); promotional strategy
Synthetic Merchant	Fake storefront with AI-generated content
Business profile	Declared products/services, business description, branding
Merchant category	E-commerce MCC (may be misclassified)
Merchant attributes	Business name, website URL, product category, location (may be synthetic)
Synthetic customers	Synthetic customers (may include legitimate-looking transactions)
Merchant devices/sessions	Device fingerprint; IP geolocation; session patterns
Synthetic transactions	Customer purchases → payment processing → non-delivery → chargebacks
Refund/chargeback events	Elevated (non-delivery)
Merchant behavioral events	Storefront creation → promotion → customer purchases → non-delivery → chargebacks → abandonment
Merchant relationships	Connections to other fake storefronts (shared infrastructure)
Network relationships	Fraud ring clustering
Temporal Behavior	Storefront creation timing; operation duration; promotional timing; abandonment patterns
GenAI-generated content	Storefront website (product galleries, team photos, testimonials, product descriptions, branding)
Attack Labels	Fake storefront; scam merchant; brand impersonation
Variant / Mutation Parameters	Product category; storefront quality; impersonation type; domain rotation; content variation; infrastructure changes; promotional strategy
Legitimate Counterpart	New e-commerce businesses; legitimate online stores
Realistic Edge Cases	Partial fraud (some goods shipped); legitimate-looking storefronts; storefronts on legitimate platforms

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
Fake storefronts generated in under 10 minutes	McAfee	13NewsNow	2025	Industry report	VERIFIED
Professional-looking websites with product galleries, team pages, testimonials	EBRAND	EBRAND	2025	Industry analysis	VERIFIED
Studio-grade product images, founder headshots, authentic-sounding reviews	EBRAND	EBRAND	2025	Industry analysis	VERIFIED
AI-generated text, imagery, and entire fraudulent websites	Netcraft	Netcraft	2025	Threat research	VERIFIED
\$442 billion lost to online scams in 2025	Global Anti-Scam Alliance	Yahoo Finance	2025	Industry report	VERIFIED
Mastercard: fake merchant activity damaging trust in digital commerce	Mastercard	Yahoo Finance	2025	Payment network	VERIFIED
20,000+ fake shop domains mapped	Malwarebytes	Malwarebytes	2026	Investigation	VERIFIED
Fake shops rose 790% in Q1 2025	Avast/Threat intelligence	Malwarebytes	2026	Threat report	VERIFIED
Visa: criminals create convincing storefronts in minutes	Visa	DigitalCommerce360	2025	Payment network report	VERIFIED
450% surge in dark-web posts discussing "AI Agent" tools for merchant fraud	Visa	DigitalCommerce360	2025	Payment network report	VERIFIED
TikTok Shop: 70 million fake products using generative AI	Mashable	Mashable	2025	News report	VERIFIED

F. PRODUCT MAPPING

Component	Implementation
Generate	Agentic generation of storefront websites with AI-generated content; Algorithmic generation of transaction sequences and refund/chargeback patterns

Component	Implementation
Defend	Website content authenticity scoring; Image forensics (CNN/Transformer); NLP for AI-generated text detection; Graph Neural Network for storefront network analysis; Refund/chargeback anomaly detection
Detection Approach	CNN/Transformer for image authenticity; NLP for AI-generated text detection; Graph Neural Network for storefront network analysis; Anomaly detection for refund/chargeback patterns
Mitigation	Allow (verified legitimate); Enhanced monitoring (watch for refund/chargeback patterns); Hold (delay onboarding pending review); Block (reject; flag in consortium intelligence)
Evaluation	Precision; Recall; AUC; False Positive Rate; Time-to-Detection

FAMILY MCH-004: AI-POWERED TRANSACTION LAUNDERING

A. IDENTITY

Field	Value
Attack ID	MCH-004
Attack Family	AI-Powered Transaction Laundering
Attack Subtypes / Variants	Transaction laundering; Commercial laundering; Front merchant laundering; Network laundering; Merchant collusion (as technique)
Primary Lifecycle Stage	Merchant (Stage 6 – Payment Acceptance)
Secondary / Cross-stage Stages	Merchant onboarding (Stage 6); Payment Initiation (Stage 5); AML/Compliance (Stage 17); Acquirer (Stage 7)
Target / Actor	Acquirer monitoring, payment processing; The front merchant serves as a laundering vehicle
Attack Objective	Process payments for illicit activity or launder money through a legitimate-appearing merchant account
Merchant Type / Context	High-volume merchants; E-commerce; Digital goods; Cross-border merchants

B. FRAUD UNDERSTANDING

What is the attack?

Attackers use AI-generated front websites and merchant profiles to create convincing front businesses that process payments for illicit activity – laundering money through the payment system. Transaction laundering has evolved, with fraudsters able to deploy AI-built “front” websites so convincing that they evade standard onboarding detection. This is the “single highest-consequence form of merchant fraud” for acquirers.

Traditional mechanism:

Before GenAI, transaction launderers created basic “front websites” with telltale signs of being fraudulent. Detection was possible through obvious red flags.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Front quality	Basic, detectable	Sophisticated sites appearing legitimate
Speed	Months to establish	Days to build and scale
Scalability	Limited	Industrial-scale
Detection evasion	Obvious red flags	Evades standard onboarding

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to generate convincing front websites that evade detection, scale front creation, and rapidly recreate fronts when detected.

HARD GenAI Removal Test result: PASS – GenAI creates a qualitatively new capability. The transformation from detectable front websites to indistinguishable laundering fronts is a qualitative leap. Scam operations previously requiring months can now be built and scaled in days.

Prerequisites:

- Access to generative AI tools (website generation, image generation, copy generation)
- Ability to register a business entity (may be synthetic)
- Ability to obtain a merchant account
- Illicit business needing payment processing
- Understanding of laundering techniques

Attack Flow:

1. Attacker identifies target industry for front business
2. Uses GenAI to generate convincing front website and business profile
3. Registers business entity (may be synthetic or real)
4. AI-generated front submitted for onboarding
5. Merchant account provisioned based on convincing facade
6. Illicit transactions processed through front merchant
7. Transactions appear legitimate (volume, velocity, amounts within expected ranges)
8. Funds laundered through shell merchant
9. Front may be abandoned and recreated

Merchant Lifecycle Point: FRAUD ENTERS AT MERCHANT ONBOARDING (the front is created for onboarding) AND CONTINUES THROUGH PAYMENT ACCEPTANCE (laundering transactions processed).

Downstream Payment / Financial Consequence:

- Funds laundered through shell merchant
- Regulatory penalties for acquirer
- Scheme violations (BRAM/VIRP)
- "Single highest-consequence form of merchant fraud" for acquirers

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Merchant Intelligence Required?
Merchant identity	Business details, registration	May be synthetic	Entity resolution	Yes
Merchant profile	Website content	AI-generated; template-heavy	CNN/Transformer; NLP	No
Transaction	Volume, velocity, patterns	Laundering indicators	ML anomaly detection	Yes
Customer relationship	Customer geography, patterns	Inconsistent with declared business	Rules + ML	No
Device	Device fingerprint	Infrastructure patterns	Device fingerprinting	Yes
Session	Session patterns	Automation indicators	Anomaly detection	No
Behavioral	Administrative behavior	Unnatural patterns	Behavioral analytics	No
Network/graph	Merchant connections	Laundering networks	Graph Neural Network	Yes — essential
Temporal	Timing patterns	Off-hours, rapid changes	Velocity rules + ML	Yes
Transaction laundering	Inconsistent descriptors	MCC vs. actual business	Rules + ML	Yes

Evasion Considerations:

An adaptive attacker might attempt to:

- Include some genuine transactions to build credibility
- Gradually increase laundering volume to avoid velocity flags
- Use varied customer geographies and amounts to appear normal
- Rotate front websites and merchant accounts
- Train AI models on real transaction data to generate "clean" patterns
- Use intricate networks of shell companies

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
Legitimate high-volume business	High transaction volume	Blocking successful merchants
New e-commerce business	New website, limited history	Blocking legitimate startups
Cross-border business	Unusual geography	Blocking legitimate international sales

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
Professional website design	"Too perfect" images	False positives on professional design

D. SIMULATION INTELLIGENCE

Field	Value
Simulation Type	Hybrid (Agentic + Algorithmic)
Generation Parameters	Industry; laundering method; volume; infrastructure patterns
Synthetic Merchant	Front business with AI-generated website and profile
Business profile	Declared product/service, website URL, business description
Merchant category	Low-risk MCC (misrepresented)
Merchant attributes	Business name, registration, location, industry (may be synthetic)
Synthetic customers	Synthetic customers (laundering transactions)
Merchant devices/sessions	Device fingerprint; IP geolocation; session patterns
Synthetic transactions	Mix of legitimate-looking and laundering transactions
Refund/chargeback events	May be low
Merchant behavioral events	Front creation → onboarding → laundering operation → potential abandonment
Merchant relationships	Connections to laundering networks
Network relationships	Shell company networks; straw signer networks
Temporal Behavior	Front creation timing; laundering duration; transaction timing patterns
GenAI-generated content	Front website; business descriptions; product listings
Attack Labels	Transaction laundering; merchant laundering; front merchant
Variant / Mutation Parameters	Industry; laundering method; volume; domain rotation; content variation; transaction pattern variation
Legitimate Counterpart	Legitimate high-volume businesses; new e-commerce businesses
Realistic Edge Cases	Partially legitimate fronts; merchant unaware of laundering

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
AI-built front websites evade standard onboarding detection	The Financial Revolutionist	The FR	2025	Industry analysis	VERIFIED
Innocuous-looking storefronts generated by AI to mask illegal payment processing	The Financial Revolutionist	The FR	2026	Industry analysis	VERIFIED
Scam operations previously requiring months can now be built and scaled in days using AI	Ballerine	The Paypers	2026	Industry report	VERIFIED
Entire e-commerce websites generated in seconds for transaction laundering	LegitScript	LegitScript	2025	Webinar analysis	VERIFIED
Synthetic merchant sites with professional compliance docs, AI-generated every element	Visa	Benzinga	2025	Payment network report	VERIFIED
Transaction laundering is the "single highest-consequence form of merchant fraud"	Onlayer	Onlayer	2026	Industry guide	VERIFIED

F. PRODUCT MAPPING

Component	Implementation
Generate	Agentic generation of front websites; Algorithmic generation of laundering transaction sequences; Network relationship generation
Defend	Website content authenticity scoring; Image forensics (CNN/Transformer); Graph Neural Network for laundering network analysis; Transaction anomaly detection; MCC consistency scoring
Detection Approach	CNN/Transformer for image authenticity; Graph Neural Network for laundering network analysis; Anomaly detection for transaction patterns; ML classification for MCC consistency
Mitigation	Allow (verified legitimate); Enhanced monitoring (watch for laundering patterns); Hold (delay onboarding; enhanced due diligence); Block (reject; flag in consortium intelligence)

Component	Implementation
Evaluation	Precision; Recall; AUC; False Positive Rate; Time-to-Detection

FAMILY MCH-005: AI-GENERATED REFUND / RETURN ABUSE

A. IDENTITY

Field	Value
Attack ID	MCH-005
Attack Family	AI-Generated Refund / Return Abuse
Attack Subtypes / Variants	AI-generated fake damage images; AI-generated damage videos; Product defect fabrication; Fresh produce/ food refund fraud; Wardrobing (with AI evidence); Organized refund rings
Primary Lifecycle Stage	Merchant (Stage 6 – Post-Transaction)
Secondary / Cross-stage Stages	Payment Initiation (Stage 5 – original purchase); Dispute/Chargeback (Stage 15 – may follow)
Target / Actor	E-commerce merchants, online retailers, payment platforms
Attack Objective	Obtain refunds without returning products, keeping both the goods and the refund
Merchant Type / Context	E-commerce; Fresh produce/food items; Consumer goods; Beauty products; Electronics

B. FRAUD UNDERSTANDING

What is the attack?

Customers use generative AI to create photorealistic fake damage images and videos to claim refunds for items they never return or that were never damaged. Global fraud detection firm Forter reports a 15% spike in AI-doctored refund claims as organized rings move to automate "damaged goods" evidence. Generative AI is democratizing e-commerce fraud, with scammers increasingly using AI-generated images to secure refunds for items they never return.

Traditional mechanism:

Before GenAI, refund fraud required physical damage to products (traceable), crude Photoshop forgeries (detectable), or exploiting policy loopholes. Detection was possible through formatting errors, lack of consistent evidence, and manual review of return patterns.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Evidence quality	Crude forgeries, physical damage	Photorealistic, indistinguishable from real
Speed	Manual, time-consuming	3 minutes to generate convincing images
Scalability	Limited	Organized crime groups automate at scale
Detection evasion	Detectable artifacts	Increasingly indistinguishable from real

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to generate photorealistic fake damage images without physical damage, create convincing evidence at scale, and evade image-based fraud detection.

HARD GenAI Removal Test result: PASS – GenAI creates a qualitatively new capability. The transformation from crude forgeries and physical damage to photorealistic fake evidence that is "indistinguishable from real structural defects" is a qualitative leap.

Prerequisites:

- Access to generative AI image/video generation tools
- Ability to make purchases and submit refund claims
- Understanding of merchant refund policies

Attack Flow:

1. Attacker identifies target product category
2. Selects generative AI tools for image/video generation
3. Makes a legitimate purchase
4. Receives goods

- 5. Uses GenAI to generate fake damage images/videos
- 6. Submits refund claim with AI-generated evidence
- 7. Refund claim appears legitimate; images difficult to distinguish from real damage
- 8. Refund authorized; customer keeps goods and receives refund
- 9. Organized rings may automate at scale using rotating IP addresses

Merchant Lifecycle Point: FRAUD ENTERS AT POST-TRANSACTION REFUND PROCESSING — the customer submits AI-generated evidence with the refund claim.

Downstream Payment / Financial Consequence:

- Direct loss to merchant (product cost + refund + shipping)
- 15% spike in AI-doctored refund claims globally
- Refund/policy abuse remains top fraud threat (45% of merchants)
- Organized crime rings automating fraud at scale
- Stricter return policies for legitimate customers
- Higher prices for all consumers

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Merchant Intelligence Required?
Transaction	High refund rate	Abuse indicator	Rules + ML	Yes
Transaction	Refund claim shortly after delivery	Timing anomaly	Rules	No
Transaction	Repeated refunds across merchants	Pattern of abuse	Cross-merchant intelligence	Yes — essential
Evidence	AI-generated image artifacts	Forgery detection	CNN/Transformer	No
Evidence	Metadata anomalies	Provenance check	Metadata analysis	No
Evidence	Inconsistent product features	Image inconsistencies	ML + Rule-based	No
Evidence	Same artifact patterns across claims	Organized fraud	Image fingerprinting	Yes
Behavioral	Customer submits only photos (no physical return)	Refund abuse pattern	Rules	No
Network	Same IP/device used for multiple claims	Fraud ring detection	Graph analytics	Yes
Temporal	Unusual refund timing	Anomaly detection	Anomaly detection	No

Evasion Considerations:

An adaptive attacker might attempt to:

- Generate increasingly realistic images with realistic noise and artifacts
- Use different AI tools/patterns across claims to avoid fingerprinting
- Make legitimate purchases occasionally to build positive history
- Distribute claims across accounts to avoid velocity flags
- Combine AI-generated evidence with partial physical damage
- Use rotating IP addresses

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
Genuine product damage	Customer submits damage photos	Blocking legitimate refund claims
Poor product quality	Customer returns damaged goods	Blocking legitimate quality complaints
Shipping damage	Customer claims item arrived damaged	Blocking legitimate shipping damage claims
Low-quality photos	Images may appear unusual	False positives on legitimate claims with poor photography
Legitimate high return rate	Customer buys/returns frequently	Blocking legitimate high-volume shoppers

D. SIMULATION INTELLIGENCE

Field	Value
Simulation Type	Algorithmic
Generation Parameters	Product category; damage type; image quality; refund amount; merchant type; timing patterns
Synthetic Merchant	N/A (customer is the attacker)
Business profile	E-commerce merchant types
Merchant category	Various MCCs
Merchant attributes	Product categories; refund policies
Synthetic customers	Customer personas (purchase history, refund history, behavior patterns)
Merchant devices/sessions	Customer device fingerprint; IP geolocation
Synthetic transactions	Purchase → Delivery → AI-generated fake evidence → Refund claim → Refund processed
Refund/chargeback events	Refund claims with AI-generated evidence
Merchant behavioral events	Normal purchase → Suspicious refund claim with AI-generated evidence
Merchant relationships	Customer-merchant relationships
Network relationships	Fraud ring connections; cross-merchant abuse patterns
Temporal Behavior	Purchase-to-refund timing; refund frequency; seasonal patterns
GenAI-generated content	AI-generated damage images (cracked screens, torn items, spoiled food, broken electronics); AI-generated damage videos
Attack Labels	Refund fraud; return abuse; policy abuse
Variant / Mutation Parameters	Product category; damage type; image quality; refund amount; merchant type; image artifact variation; timing randomization
Legitimate Counterpart	Genuine product damage claims; legitimate returns; shipping damage claims
Realistic Edge Cases	Partial fraud (some genuine damage, some fabricated); mixed claims; dispute escalation

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
AI-generated damage photos driving sharp rise in fraudulent refund claims	Jing Daily	Jing Daily	2025	News report	VERIFIED
AI-generated refund fraud targets fresh produce and food items	China Daily	China Daily	2025	News report	VERIFIED
Customers use AI to fake product damage, edited images look convincing	Times Now	Times Now	2025	News report	VERIFIED
"Open AI tool, input commands, 3 minutes to generate convincing quality issue images"	Workers' Daily	CE.CN	2025	News report	VERIFIED
AI-generated images "indistinguishable from real structural defects"	Industry analysis	Pass B	2025	Industry analysis	REASONABLE INFERENCE
15% spike in AI-doctored refund claims globally	Forter	Industry analysis	2025	Industry analysis	VERIFIED
Refund abuse: 37% of businesses affected	Ravelin	Startups.co.uk	2026	Industry report	VERIFIED
Organized crime groups using tools to automate fraud at scale	Industry analysis	Fraud/security industry	2025	Industry report	VERIFIED

F. PRODUCT MAPPING

Component	Implementation
Generate	Algorithmic generation of AI-generated damage images with varying quality levels; Parameterization by product category, damage type, image quality; Refund claim sequences
Defend	CNN/Transformer for image forgery detection; Anomaly detection for refund patterns; Graph analytics for cross-account pattern detection; Metadata analysis for image provenance
Detection Approach	CNN/Transformer for image authenticity scoring; Anomaly detection for refund behavior; Graph analytics for organized fraud detection; Rules for refund velocity

Component	Implementation
Mitigation	Allow (low risk, genuine images, normal pattern); Challenge (medium risk – request additional evidence, video verification); Hold (high risk – delay refund, manual review); Block (confirmed fraud – deny claim, account review)
Evaluation	Precision; Recall; AUC; False Positive Rate; Time-to-Detection

FAMILY MCH-006: AI-POWERED MERCHANT SOCIAL ENGINEERING

A. IDENTITY

Field	Value
Attack ID	MCH-006
Attack Family	AI-Powered Merchant Social Engineering
Attack Subtypes / Variants	Merchant phishing; Deepfake impersonation of bank/platform; Voice cloning attacks; Merchant credential phishing; Customer impersonation; Merchant social engineering campaigns
Primary Lifecycle Stage	Merchant (Stage 6 – Account Management / Authentication)
Secondary / Cross-stage Stages	Authentication (Stage 4 – credential compromise); Device/Session (Stage 3 – session anomalies); Payment Initiation (Stage 5 – post-compromise fraud)
Target / Actor	Merchants, merchant employees, merchant customers
Attack Objective	Steal merchant login credentials, API keys, or banking details; manipulate merchants into authorizing fraudulent payments; or impersonate merchants to defraud customers
Merchant Type / Context	All merchant types; particularly small to medium businesses with limited security awareness

B. FRAUD UNDERSTANDING

What is the attack?

Attackers use generative AI – deepfakes, voice cloning, and LLM-generated personalized content – to target merchants through social engineering, stealing credentials, payment data, or manipulating merchants into authorizing fraudulent transactions. Deepfake impersonation is a key risk. AI-generated phishing emails remain a core tactic. Scammers are using AI to impersonate customers, colleagues, and family.

Traditional mechanism:

Before GenAI, social engineering relied on generic phishing emails (detectable by poor grammar) and human callers (limited scalability). Detection was possible through language errors and impersonation quality.

GenAI transformation:

Dimension	Pre-GenAI	GenAI-Transformed
Content quality	Generic, detectable errors	Hyper-personalized, grammatically perfect
Personalization	Manual, limited	AI-generated personalized content referencing real details
Voice	Human impersonation	Voice clones indistinguishable from real
Video	None or pre-recorded	Real-time deepfake video impersonation
Scalability	Human-limited	Industrial-scale, automated

Why GenAI is load-bearing:

If GenAI is removed, the attacker loses the ability to generate grammatically perfect, personalized phishing at scale, create convincing deepfake videos and voice clones, hyper-personalize attacks based on merchant intelligence, and adapt social engineering in real time.

HARD GenAI Removal Test result: PASS – GenAI creates a qualitatively new capability. The transformation from generic, detectable social engineering to hyper-personalized, multi-channel deception at scale is a qualitative leap.

Prerequisites:

- Access to generative AI tools (LLMs, voice cloning, deepfake video)
- Merchant contact information or public data for personalization
- Communication channel (email, phone, messaging)

Attack Flow:

1. Attacker gathers intelligence on target merchant (OSINT, social media, public data)
2. Selects GenAI tools for content generation
3. Attacker contacts merchant via email, phone, or messaging
4. Uses AI-generated content to impersonate trusted entity (bank, platform, customer, colleague)
5. Merchant is manipulated into providing credentials or authorizing transactions
6. Credentials harvested; account compromised
7. Fraudulent transactions processed through merchant account
8. Funds stolen or customer data exposed

Merchant Lifecycle Point: FRAUD ENTERS AT ACCOUNT MANAGEMENT – the merchant (or employee) is targeted post-onboarding through social engineering.

Downstream Payment / Financial Consequence:

- Merchant account compromised
- Funds stolen or fraudulent transactions processed
- Customer data exposed
- Reputational damage to merchant and payment ecosystem
- 3 in 10 fraud attempts targeting retailers are AI-generated

C. DETECTION INTELLIGENCE

Observable Signals:

Signal Category	Observable Signal	Why It Matters	Detection Method	Cross-Merchant Intelligence Required?
Merchant identity	N/A (merchant is legitimate victim)	N/A	N/A	No
Merchant profile	N/A	N/A	N/A	No
Transaction	Unusual amounts/beneficiaries	Post-compromise fraud	ML anomaly detection	Yes
Customer relationship	N/A	N/A	N/A	No
Device	New devices; unusual locations	Credential compromise	Device fingerprinting	No
Session	Unusual session patterns	Account takeover	Session anomaly detection	No
Behavioral	Unusual administrative behavior	Compromise indicator	Behavioral analytics	No
Network/graph	Fraud connections	Fraud network	Graph Neural Network	Yes
Temporal	Unusual timing	Anomaly detection	Temporal anomaly detection	No
Communication	Phishing emails; deepfake content	Social engineering	NLP; Deepfake detection	Yes – campaign identification

Evasion Considerations:

An adaptive attacker might attempt to:

- Use increasingly sophisticated deepfakes to eliminate artifacts
- Personalize communications with real merchant details
- Use multiple communication channels to build credibility
- Adapt scripts based on merchant responses in real time
- Target merchants less likely to detect fraud (small businesses, busy owners)
- Use legitimate platforms for communication

Legitimate Behavior That Could Look Similar:

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
Legitimate merchant credential change	Merchant updates password	Blocking legitimate credential changes
Legitimate bank communication	Professional banking communication	Flagging genuine bank communications
Legitimate new device	Merchant uses new device	Blocking legitimate device changes
Legitimate large transaction	Merchant processes large payment	Blocking legitimate business transactions

D. SIMULATION INTELLIGENCE

Field	Value
Simulation Type	Agentic
Generation Parameters	Attack channel (email/phone/video); impersonation type; personalization level; content variation; timing variation
Synthetic Merchant	Merchant personas (legitimate merchants with transaction histories)
Business profile	Products/services, transaction patterns
Merchant category	Various MCCs
Merchant attributes	Business type, size, industry, location
Synthetic customers	N/A
Merchant devices/sessions	Device fingerprint; IP geolocation; session patterns
Synthetic transactions	Post-compromise fraudulent transactions
Refund/chargeback events	May increase post-compromise
Merchant behavioral events	Normal operation → Phishing contact → Credential compromise → Unusual administrative behavior → Fraudulent transactions
Merchant relationships	N/A
Network relationships	Fraud network connections
Temporal Behavior	Phishing timing; compromise timing; fraud timing
GenAI-generated content	Phishing emails; deepfake videos; voice clones; personalized social engineering content
Attack Labels	Merchant phishing; merchant social engineering; deepfake impersonation
Variant / Mutation Parameters	Attack channel (email/phone/video); impersonation type; personalization level; content variation; timing variation
Legitimate Counterpart	Legitimate bank communications; legitimate credential changes
Realistic Edge Cases	Partial compromise; multi-channel attacks

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
GenAI enables more effective social engineering, deepfakes, voice clones	Mastercard	Mastercard	2025	Payment network	VERIFIED
AI scams 2.0 blends social engineering with automated content creation	SecurityBrief	SecurityBrief	2025	Industry analysis	VERIFIED
Deepfake support attacks targeting merchants	Ecommpay	Ecommpay	2025	Industry report	VERIFIED
AI-powered shopping tools lack anti-scam protections	F-Secure	F-Secure	2025	Security research	VERIFIED
3 in 10 fraud attempts targeting retailers are AI-generated	Industry report	Fraud/security industry	2025	Industry report	VERIFIED
Merchant phishing identified as fraud vector	Razorpay	Razorpay	2025	Industry guide	VERIFIED

F. PRODUCT MAPPING

Component	Implementation
Generate	Agentic generation of phishing content (LLM); Deepfake video generation; Voice clone generation; Personalized social engineering scenarios
Defend	NLP for AI-generated content detection; Deepfake detection (CNN/Transformer); Behavioral anomaly detection; Cross-merchant campaign identification
Detection Approach	NLP for AI-generated content; Deepfake detection; Behavioral anomaly detection; Graph Neural Network for campaign identification
Mitigation	Allow (normal activity); Challenge (step-up verification for suspicious communications); Hold (delay transactions pending verification); Block (suspend account; customer notification)
Evaluation	Precision; Recall; AUC; False Positive Rate; Time-to-Detection

PART 4 – FINAL COVERAGE AUDIT

Merchant Attack Surface	Considered?	Covered By	Separate Family?	Reason
Merchant onboarding	✓	MCH-001, MCH-002, MCH-003, MCH-004	No	Covered across multiple families
Merchant identity fraud	✓	MCH-001	Yes	Synthetic merchant identity creation
Fraudulent merchant creation	✓	MCH-001	No	Variant of synthetic onboarding
Synthetic/fictitious merchant	✓	MCH-001	No	Variant of synthetic onboarding
Merchant impersonation	✓	MCH-001	No	Variant – uses same GenAI capabilities
Merchant account takeover	✓	MOVED → Authentication	No	ATO belongs to Authentication
Merchant credential abuse	✓	MCH-003	No	Enables fake storefront operations
Merchant administrator compromise	✓	MCH-006	No	Social engineering targeting administrators
Merchant configuration manipulation	✓	MCH-003	No	Post-compromise activity
Fake merchant storefront	✓	MCH-003	Yes	Distinct family – consumer-facing scam stores
Fraudulent merchant website	✓	MCH-003	No	Variant of fake storefronts
Checkout manipulation	✓	MCH-003	No	Fake storefronts manipulate checkout
Payment-page manipulation	✓	MCH-003	No	Fake storefronts manipulate payment pages
Merchant-side payment redirection	✓	MCH-004	No	Transaction laundering variant
Transaction laundering	✓	MCH-004	Yes	"Highest-consequence" merchant fraud
Merchant/customer collusion	✓	MCH-002, MCH-004	No	Collusion enables laundering/shell fraud
Merchant collusion	✓	MCH-002, MCH-004	No	Collusion enables laundering/shell fraud
Related merchant networks	✓	MCH-001, MCH-002, MCH-003, MCH-004	No	Detected via graph analytics
Merchant behavioral anomalies	✓	All families	No	Detection signal across all families
Refund manipulation	✓	MCH-005	Yes	Distinct family – AI-generated fake evidence
Merchant-side transaction abuse	✓	MCH-001, MCH-002, MCH-003, MCH-004	No	Covered across multiple families
Merchant reputation manipulation	✓	MCH-003	No	Fake reviews are part of fake storefronts
Fake reviews/reputation	✓	MCH-003	No	AI-generated reviews part of fake storefronts
AI-generated merchant content	✓	MCH-002, MCH-003	No	Content generation is the enabler
AI-generated merchant/customer interaction	✓	MCH-006	No	Social engineering interactions
Merchant automation	✓	MCH-002, MCH-003, MCH-004	No	Automation is the enabler
Merchant support impersonation	✓	MCH-006	No	Social engineering variant
Merchant collateral	✓	MCH-001, MCH-002, MCH-003, MCH-004	No	Collateral creation is the enabler
Merchant data manipulation	✓	MCH-001	No	Synthetic data creation
Merchant resale	✓	MCH-001	No	Downstream activity
Merchant verification bypass	✓	MCH-001	No	Onboarding bypass
Merchant account farming	✓	MCH-001	No	Mass account creation
Merchant bust-out	✓	MCH-001, MCH-004	No	Downstream activity
Merchant money mule	✓	MCH-004	No	Transaction laundering

Excluded Candidates Explained

Candidate	Decision	Reason
Merchant Account Takeover	MOVED → Authentication	ATO belongs to Authentication; merchant-specific context doesn't make it distinct
Chargeback Fraud / Friendly Fraud	MOVED → Dispute/Chargeback	Primary mechanism is dispute process
Credit Card Testing via Merchant	DROPPED	GenAI amplification, not transformation
Merchant Enumeration	DROPPED	GenAI amplification, not transformation
Merchant Credit Fraud	DROPPED	GenAI amplification; overlaps with AML/Compliance
Transaction Structuring	DROPPED	GenAI amplification; overlaps with AML/Compliance
Merchant Insolvency Fraud	DROPPED	GenAI not load-bearing
Merchant Bankruptcy Fraud	DROPPED	GenAI not load-bearing
Merchant Tax Evasion	DROPPED	GenAI not load-bearing
Merchant Insurance Fraud	DROPPED	GenAI not load-bearing
Merchant Collateral Fraud	DROPPED	GenAI not load-bearing
POS Manipulation (Physical)	DROPPED	GenAI not load-bearing
In-person Merchant Fraud	DROPPED	GenAI not load-bearing
Merchant Supply Chain Fraud	DROPPED	GenAI not load-bearing
Merchant Warranty Fraud	DROPPED	GenAI not load-bearing
Merchant Inventory Fraud	DROPPED	GenAI not load-bearing
Merchant Pricing Fraud	DROPPED	GenAI not load-bearing
Merchant Promotional Abuse	DROPPED	GenAI amplification, not transformation
Merchant Loyalty Fraud	DROPPED	GenAI amplification, not transformation
Merchant Gift Card Fraud	DROPPED	GenAI amplification, not transformation
Merchant Coupon Abuse	DROPPED	GenAI amplification, not transformation
Merchant Referral Abuse	DROPPED	GenAI amplification, not transformation
Merchant Affiliate Fraud	DROPPED	GenAI amplification, not transformation
Merchant Drop-shipping Fraud	DROPPED	Covered by MCH-002/003; not distinct
Merchant Fulfillment Fraud	DROPPED	Covered by MCH-003; not distinct

PART 5 – FINAL SUMMARY

Table 1 – FINAL MERCHANT ATTACK TAXONOMY

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	Reason for Distinctness
MCH-001	AI-Generated Synthetic Merchant Onboarding	Synthetic business fraud; Merchant impersonation; Merchant misrepresentation/MCC fraud; Synthetic UBO/director fraud	PASS	Hybrid	Merchant	GenAI enables industrial-scale synthetic merchant creation with complete, internally consistent business files – KYB's cross-document consistency check is a test generated files pass by construction
MCH-002	AI-Generated Shell/Facade Merchant Fraud	Front merchant fraud; Transaction laundering fronts; Shell company networks	PASS	Hybrid	Merchant	GenAI enables convincing front websites in minutes that evade standard onboarding detection – "single highest-consequence form of merchant fraud" for acquirers
MCH-003	AI-Generated Fake Merchant Storefronts	Scam merchant storefronts; Brand impersonation; Triangulation fraud; Credential/API abuse	PASS	Hybrid	Merchant	GenAI enables professional-grade fake storefronts with studio-quality content, generated in minutes – scammers no longer look like scammers

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	Reason for Distinctness
MCH-004	AI-Powered Transaction Laundering	Commercial laundering; Network laundering; Front merchant laundering	PASS	Hybrid	Merchant	GenAI enables AI-built front websites that evade standard onboarding detection – scam operations previously requiring months can now be built and scaled in days
MCH-005	AI-Generated Refund/Return Abuse	AI-generated fake damage images; AI-generated damage videos; Product defect fabrication; Organized refund rings	PASS	Algorithmic	Merchant	GenAI enables photorealistic fake damage images in minutes, "indistinguishable from real structural defects" – qualitative leap from physical damage or crude forgeries
MCH-006	AI-Powered Merchant Social Engineering	Merchant phishing; Deepfake impersonation; Voice cloning attacks; Customer impersonation	PASS	Agentic	Merchant	GenAI enables hyper-personalized, multi-channel social engineering at scale – qualitative leap from generic, detectable phishing

Total: 6 final Merchant attack families

Total variants: 18+ (across 6 families)

Table 2 – ATTACK × OBSERVABLE SIGNAL MATRIX

Attack Family	Merchant Identity	Merchant Profile	Transaction	Customer Relationship	Device	Behavioral	Network	Temporal	Refun
MCH-001: Synthetic Merchant Onboarding	Fabricated/ inconsistent business details	Inconsistent MCC, business type	Anomalous volume/velocity	Inconsistent with declared business	Automation patterns	Unnatural form-filling	Synthetic clusters	Off-hours; rapid succession	N/A
MCH-002: Shell/Facade Merchant Fraud	May be synthetic/ inconsistent	Template-heavy, AI-generated website	Laundering patterns	Inconsistent with declared business	Infrastructure patterns	Unnatural administrative behavior	Shell networks	Off-hours; rapid changes	May be
MCH-003: Fake Merchant Storefronts	May be inconsistent	AI-generated content artifacts	High refund/ chargeback rates	Broad/inconsistent geography	Infrastructure patterns	Administrative anomalies	Storefront networks	Rapid creation/ abandonment	High r charge
MCH-004: Transaction Laundering	May be synthetic	AI-generated front content	Laundering indicators	Inconsistent with declared business	Infrastructure patterns	Unnatural administrative behavior	Laundering networks	Off-hours; rapid changes	May be
MCH-005: Refund/Return Abuse	Legitimate customer identity	N/A	High refund rate; claim shortly after delivery	N/A	Device fingerprinting	Claim patterns	Cross-merchant abuse patterns	Unusual refund timing	High r charge
MCH-006: Merchant Social Engineering	N/A (victim is legitimate)	N/A	Post-compromise anomalies	N/A	New devices; unusual locations	Unusual administrative behavior	Fraud campaign connections	Unusual timing	May in

Table 3 – ATTACK × SIMULATION MATRIX

Attack Family	Synthetic Merchant	Business Profile	Merchant Category	Synthetic Customers	Transactions	Behavioral Events	Network	Content	Agentic/I
MCH-001	Fictitious business with fabricated structure	Declared revenue, employees, years in operation	Low-risk MCC	Synthetic customers (for laundering)	Cultivation → fraud/ laundering	Onboarding flow (form filling, document upload)	Synthetic clusters	AI-generated business documents	Hybrid
MCH-002	Shell/facade business with front website	Declared product/service, website URL	Low-risk MCC	Synthetic customers (legitimate-looking)	Mix of legitimate-looking and laundering	Shell creation → onboarding → laundering → abandonment	Shell networks	AI-generated front websites	Hybrid
MCH-003	Fake storefront with AI-generated content	Declared products/services, branding	E-commerce MCC	Synthetic customers (purchasing)	Purchases → non-delivery → chargebacks	Storefront creation → promotion → abandonment	Storefront networks	AI-generated storefronts, product images, reviews	Hybrid

Attack Family	Synthetic Merchant	Business Profile	Merchant Category	Synthetic Customers	Transactions	Behavioral Events	Network	Content	Agentic//
MCH-004	Front business with AI-generated website	Declared product/service, website URL	Low-risk MCC (misrepresented)	Synthetic customers (laundering)	Mix of legitimate-looking and laundering	Front creation → onboarding → laundering	Laundering networks	AI-generated front websites	Hybrid
MCH-005	N/A (customer is attacker)	N/A	Various	Customer personas with purchase/refund history	Purchase → fake damage claim → refund	Normal purchase → suspicious refund claim	Customer-merchant; cross-merchant	AI-generated damage images/videos	Algorithm
MCH-006	Merchant personas (legitimate victims)	Products/services, transaction patterns	Various	N/A	Post-compromise fraudulent transactions	Normal → phishing contact → compromise → fraud	Fraud campaign connections	AI-generated phishing, deepfakes, voice clones	Agentic

Table 4 – TAXONOMY DECISIONS

Candidate	Decision	Reason
AI-Generated Synthetic Merchant Onboarding	KEEP	Distinct family – industrial-scale synthetic merchant creation
Synthetic Merchant Identity	MERGE → MCH-001	Variant – synthetic identity is the enabler
Merchant Impersonation	MERGE → MCH-001	Variant – uses same GenAI capabilities
Merchant Misrepresentation / MCC Fraud	MERGE → MCH-001	Variant – misrepresentation is a use case
AI-Generated Shell/Facade Merchant Fraud	KEEP	Distinct family – convincing front businesses for laundering
Merchant Collusion	MERGE → MCH-002	Variant – collusion enables laundering
AI-Generated Fake Merchant Storefronts	KEEP	Distinct family – professional fake stores for consumer fraud
Merchant Credential / API Key Abuse	MERGE → MCH-003	Variant – enabler for fake storefront operations
Triangulation Fraud	MERGE → MCH-003	Variant – uses fake storefronts
Merchant Phishing	MERGE → MCH-006	Variant – social engineering targeting merchants
AI-Powered Transaction Laundering	KEEP	Distinct family – "highest-consequence" merchant fraud
AI-Generated Refund/Return Abuse	KEEP	Distinct family – photorealistic fake evidence
Chargeback Fraud / Friendly Fraud	MOVE → Dispute/Chargeback	Primary mechanism is dispute process
Merchant Account Takeover	MOVE → Authentication	ATO belongs to Authentication

Final Metrics

Metric	Value
Total final attack families	6
Total variants	18+
Important candidates merged	Synthetic Merchant Identity → MCH-001; Merchant Impersonation → MCH-001; Merchant Misrepresentation → MCH-001; Merchant Collusion → MCH-002; Merchant Credential/API Abuse → MCH-003; Triangulation Fraud → MCH-003; Merchant Phishing → MCH-006
Important candidates moved	Merchant Account Takeover → Authentication (Stage 4); Chargeback Fraud → Dispute/Chargeback (Stage 15)
Important candidates dropped	Credit Card Testing; Merchant Enumeration; Merchant Credit Fraud; Transaction Structuring; Merchant Insolvency Fraud; Merchant Bankruptcy Fraud; Merchant Tax Evasion; Merchant Insurance Fraud; POS Manipulation; In-person Merchant Fraud; Supply Chain Fraud; Warranty Fraud; Inventory Fraud; Pricing Fraud; Promotional Abuse; Loyalty Fraud; Gift Card Fraud; Coupon Abuse; Referral Abuse; Affiliate Fraud
Theoretical/future threats	AI-powered autonomous merchant agents; AI-generated merchant networks at scale; Self-evolving merchant fraud systems
Remaining research gaps	Real-world frequency of AI-generated shell merchant fraud; Specific detection evasion effectiveness against production acquirer systems; Impact of emerging GenAI architectures on KYB verification; Effectiveness of cross-merchant intelligence sharing
Merchant contexts covered	Onboarding; Account management; Payment acceptance; Post-transaction; All merchant types; E-commerce; Physical retail; Digital goods; High-volume merchants; Cross-border merchants

Metric	Value
Important Merchant surfaces not represented as separate families	Account takeover (moved to Authentication); Chargeback fraud (moved to Dispute/Chargeback); Various dropped due to GenAI not being load-bearing

Unifying Theme

The unifying theme across all 6 Merchant families is the **shift from detectable to indistinguishable** — fraudsters no longer rely on crude forgeries, poor-quality storefronts, or detectable social engineering; they use GenAI to generate professional-grade merchant fraud that appears legitimate to consumers, payment systems, and verification processes. The tell-tale signs of a scam used to be obvious; today's generative AI removes those cues. Scammers no longer look like scammers.